

HARRISON JANSMA

Senior Applied Scientist • Generative AI & Applied ML Solutions

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SUMMARY

Senior applied scientist / ML engineer with 6+ years taking applied AI from research to production for enterprise customers — generative AI, real-time NLP, and large-scale ML systems with measurable business impact. Specialized in agentic LLM workflows, retrieval-augmented generation, low-latency serving, and rigorous evaluation. Proven record translating ambiguous business problems into shipped models serving thousands of concurrent users and leading cross-functional teams.

CORE SKILLS

Applied Science & GenAI: LLM orchestration, agentic / multi-step reasoning workflows, prompt engineering, retrieval-augmented generation (RAG), fine-tuning, transformers (DistilBERT/BERT), embeddings, NLU, sentiment & emotion classification, reinforcement learning, simulation

ML at Scale: PyTorch, TensorFlow, Hugging Face, scikit-learn; streaming inference (Apache Kafka, AWS Lambda, DynamoDB); vector databases; low-latency serving (200+ TPS @ ~100 ms); AWS (S3), Snowflake

Evaluation & Observability: LLM-as-a-judge, offline & online evaluation, quality-metric design, governance dashboards, New Relic, PagerDuty, Splunk

Business Impact: Credit-decisioning models (\$32M+/yr value), agentic GenAI (\$1.3M/yr savings), customer-facing solution delivery

Languages & Data: Python, SQL, PySpark, Hive, Tableau

EXPERIENCE

Capital One — Plano, TX • 2020–Present

Data Science Manager (2024–Present) • Principal Data Scientist (2022–2024) • Sr. Associate, Data Science (2021–2022) • Data Process Manager (2020–2021)

Applied Generative AI (Current Focus)

- Lead Generative AI for Capital One's contact centers — applied LLM/agentic systems that augment thousands of servicing agents live, mid-conversation, with measurable business outcomes.
- Built an agentic GPT reasoning pipeline (Kafka) that auto-drafts agent notes, cutting handling time by 40 seconds and driving **\$1.3M in annual cost reduction**; added an LLM "agent-as-a-judge" review stage, reaching **90% accuracy**.
- Designed a retrieval-augmented generation system surfacing the right documents mid-task via vector-embedding retrieval over live transcripts (Kafka, AWS Lambda, vector database) — **80% document recall** in real time.
- Shipped an LLM complaint-detection system on live transcripts, raising detection from 30% (legacy ML) to **80%**.

Applied ML at Scale

- Architected Capital One's first real-time AI system for detecting frustrated customers, monitoring tens of thousands of live calls per day and alerting within ~1 second.
- Engineered parallel real-time inference pipelines sustaining **200+ transactions/sec at ~100 ms latency** (Kafka, DynamoDB, S3, Snowflake).
- Trained and deployed DistilBERT and LLM models (PyTorch, Hugging Face, scikit-learn) for streaming sentiment and NLU.

Applied Science in Credit & Leadership

- Engineered Capital One's auto-loan underwriting ML suite, decisioning **billions of dollars annually**; built a model worth **\$32M+/yr**; built a simulation engine for RL-based credit-policy optimization.
- Lead a team of ML engineers and data scientists, setting technical objectives with VP- and EVP-level stakeholders.

Sprint — Irving, TX • 2019

Data Science Intern

- Designed a model-training framework for anomaly detection on customer billing data — 5x increase in detectable failure types; only extension across an 80+ intern class.

University of Pennsylvania (via Upwork) — Remote • 2018–2019

Machine Learning Researcher — NLP

- Built an emotion-classification model and processed millions of public social-media profiles for a population-level NLP study.

EDUCATION

M.S. Computer Science (Data Science) — The University of Texas at Dallas • 2018–2020

B.B.A. Mathematics, Business Fellows — Baylor University • 2013–2017 — Magna Cum Laude, GPA 3.86; Honors Program.